

ReadMe: monte_carlo_gp_assignment.ipynb

Step 1 — Imports and Paths: All required libraries (GeoPandas, NumPy, Matplotlib) and file paths are defined using relative paths so the notebook runs for any collaborator who clones the repository without editing paths. Key constants are defined here including `N_SIMULATIONS = 1000`, `RURAL_DISPLACEMENT = 5000m`, `URBAN_DISPLACEMENT = 2000m`, and `PILOT_DISTRICT = "099"` (Ganganagar, Rajasthan). The pipeline is first validated on a single district before scaling up.

Step 2 — Load and Prepare Ganganagar Village Polygons: Rajasthan village polygons are loaded and filtered to Ganganagar district (`pc11_d_id = "099"`), yielding 3,034 village polygons. These are reprojected to UTM Zone 44N (`EPSG:32644`) for accurate distance calculations in metres.

Step 3 — Assign GP Codes to Ganganagar Villages: The SHRUG-LGD crosswalk is loaded and GP LGD codes are assigned to each village using the same `shrid2` composite key approach as in `cluster_GPs_cleaning.ipynb`. GP coverage for Ganganagar is 93.6% (2,840 of 3,034 villages), yielding 343 unique GPs.

Step 4 — Dissolve Village Polygons to GP Level: Village polygons sharing the same `gp_lgd_code` are merged into GP-level boundary polygons using GeoPandas `dissolve`. This produces 343 GP polygons for Ganganagar and is the key new step that enables the Monte Carlo — we need GP boundaries to check which GP each simulated point falls in. Invalid geometries are checked and fixed if present.

Step 5 — Load and Filter NFHS Clusters to Ganganagar: NFHS4 and NFHS5 GPS cluster shapefiles are loaded, standardised, and filtered to Ganganagar using the `DHSREGNA` district name field from the DHS file. This approach is more reliable than a spatial convex hull filter, which was found to incorrectly include clusters from neighbouring Hanumangarh district. NFHS4 yields 43 clusters (31 rural) and NFHS5 yields 45 clusters (33 rural) in Ganganagar.

Step 6 — Monte Carlo GP Assignment: For each cluster, 1,000 random displacements are simulated using the actual DHS displacement distribution per specification: `angle ~ Uniform[0, 2π)` and `distance ~ Uniform[0, R]` where `R = 5km` rural, `2km` urban. This distribution is not uniform over the disk area — points closer to the displaced GPS are more likely — which is why area-based calculation alone would be a less accurate approximation. For each simulated true location, the code checks which GP polygon it falls within using a spatial join. The fraction of simulations landing in each GP gives the estimated probability that the cluster's true location is in that GP.

Step 7 — Summary Statistics: Key statistics are computed for rural clusters by survey round, including mean and median `P(primary GP)`, share of clusters above probability thresholds (`P > 0.8`, `P > 0.5`), mean number of GPs hit per cluster, and the distribution of `P(primary GP)` across probability bins.

Step 8 — District-Level Visualisation: A side-by-side map of Ganganagar is produced for NFHS4 and NFHS5, showing GP boundaries with clusters coloured by **P(primary GP)** on a red-yellow-green scale. This provides an overview of spatial uncertainty across the district.

Step 9 — Zoomed-In Cluster Detail Maps: Individual zoomed-in maps are produced for three hand-check clusters spanning the confidence range: high (**P = 68.9%**), medium (**P = 36.3%**), and low/border (**P = 24.9%**). Each map shows the cluster GPS point, the 5km displacement radius circle, GP boundary polygons, and GP names labelled with their Monte Carlo probabilities. These maps enable visual verification that the probability distributions make geographic sense before scaling up.

Step 10 — Hand-Check Text Output and Save: Detailed probability distributions are printed for the three sample clusters. Results are saved to `../data/` as **ganganagar_mc_nfhs4.csv** and **ganganagar_mc_nfhs5.csv**.

Key Outputs and Takeaways:

The Monte Carlo confirms and quantifies the findings from **cluster_GPs_cleaning.ipynb** with actual probabilities rather than binary flags. For rural clusters in Ganganagar, the mean **P(primary GP)** is approximately 0.37 for both NFHS4 and NFHS5 — meaning on average only 37% of simulated true locations fall in the assigned GP, with each cluster's displacement radius touching 7-8 distinct GPs on average. No rural cluster achieves **P > 0.8** and almost all fall in the 25-50% range.

Visual hand-checking of three sample clusters confirmed the code is working correctly: the high-confidence cluster (**P = 68.9%**) shows the GPS point sitting clearly inside a large GP that dominates the circle; the medium-confidence cluster (**P = 36.3%**) shows the GPS point sitting at the junction of several similarly-sized GPs; and the low-confidence border cluster (**P = 24.9%**) shows the displacement circle extending off the eastern edge of the district into neighbouring Hanumangarh, explaining the 604 simulations outside any GP polygon.

These results do not necessarily mean GP assignment is fatally unreliable. The critical next step is merging GP election reservation status data — once treatment status (**reserved/unreserved**) is known for each GP, it will be possible to compute **P(correct treatment status)** rather than just **P(primary GP)**. If adjacent GPs within the displacement radius share the same reservation status, spatial uncertainty in GP assignment does not translate into uncertainty in treatment assignment. This is the key quantity needed before scaling the Monte Carlo pipeline to all Rajasthan and Uttar Pradesh districts.